Straight Line: Questions and Full Solutions

Question 01

English

Let O be the origin. A point P, different from O, moves in the plane in such a way that the circle drawn on OP as a diameter always touches the line $x + y = 4$. Then the length of the latus rectum of the locus of P is equal to

(A) $4\sqrt{2}$
(B) 8
(C) $8\sqrt{2}$
(D) $16\sqrt{2}$

Only one option is correct.

हिन्दी

माना O मूल बिन्दु है। एक बिन्दु P, जो O से भिन्न है, समतल में इस प्रकार गति करता है कि OP को व्यास मानकर खींचा गया वृत्त सदैव रेखा $x + y = 4$ को स्पर्श करता है। तब P के बिन्दुपथ के नाभिलम्ब की लम्बाई बराबर है

(A) $4\sqrt{2}$
(B) 8
(C) $8\sqrt{2}$
(D) $16\sqrt{2}$

केवल एक विकल्प सही है।

Solution

Let $P = (h, k)$. The circle with OP as a diameter has centre $M\left(\frac{h}{2}, \frac{k}{2}\right)$, the midpoint of OP, and radius $\frac{1}{2}OP = \frac{1}{2}\sqrt{h^2 + k^2}$.

A circle touches a line exactly when the distance of its centre from the line equals its radius:

$$\frac{\left|\frac{h}{2} + \frac{k}{2} - 4\right|}{\sqrt{2}} = \frac{1}{2}\sqrt{h^2 + k^2}.$$

Multiplying by 2: $\dfrac{|h + k - 8|}{\sqrt{2}} = \sqrt{h^2 + k^2}$.

The left side is the distance of P from the line $x + y = 8$ and the right side is OP. So every position of P is equidistant from the point O and the line $x + y = 8$, and conversely every such point gives a touching circle.

Hence the locus is a parabola with focus $O(0, 0)$ and directrix $x + y = 8$ (its axis is the line $y = x$, which is not parallel to a coordinate axis). Squaring gives the equation $(x-y)^2+16(x+y)-64 = 0$.

For a parabola, the semi-latus rectum equals the distance of the focus from the directrix, so the latus rectum is twice this distance.

Distance of O from $x + y = 8$ is $\dfrac{8}{\sqrt{2}} = 4\sqrt{2}$, so the latus rectum is $8\sqrt{2}$.

Check: the chord through O perpendicular to the axis is along $y = -x$; on it $P = (s, -s)$ has $OP = \sqrt{2}|s|$ and distance $\dfrac{8}{\sqrt{2}}$ from the directrix, so $|s| = 4$ and the chord has length $2 \cdot 4\sqrt{2} = 8\sqrt{2}$.

Correct Option: (C)

Question 02

English

A circle C touches each of the lines $3x - 4y + 5 = 0$ and $6x - 8y - 30 = 0$, cuts off a chord of length $2\sqrt{3}$ on the x-axis and meets the y-axis. Then the length of the chord that C cuts off on the y-axis is

(A) $\dfrac{2\sqrt{35}}{3}$

(B) $\dfrac{\sqrt{35}}{3}$

(C) $\dfrac{2\sqrt{5}}{3}$

(D) $2\sqrt{3}$

Only one option is correct.

हिन्दी

एक वृत्त C प्रत्येक रेखा $3x - 4y + 5 = 0$ तथा $6x - 8y - 30 = 0$ को स्पर्श करता है, x-अक्ष पर $2\sqrt{3}$ लम्बाई की जीवा काटता है तथा y-अक्ष से मिलता है। तब C द्वारा y-अक्ष पर काटी गई जीवा की लम्बाई है

(A) $\dfrac{2\sqrt{35}}{3}$

(B) $\dfrac{\sqrt{35}}{3}$

(C) $\dfrac{2\sqrt{5}}{3}$

(D) $2\sqrt{3}$

केवल एक विकल्प सही है।

Solution

Divide the second line by 2: $3x - 4y - 15 = 0$. It is parallel to $3x - 4y + 5 = 0$, so the circle lies between two parallel lines and touches both.

Distance between the parallel lines $= \dfrac{|5 - (-15)|}{\sqrt{9 + 16}} = 4$. A circle touching both has this distance as a diameter, so $r = 2$.

Its centre (h, k) is equidistant from the two lines, so it lies on the mid-parallel $3x - 4y + \dfrac{5 - 15}{2} = 0$, i.e. $3x - 4y - 5 = 0$.

The chord on the x-axis has length $2\sqrt{r^2 - k^2}$ (here $|k|$ is the distance of the centre from the x-axis). So $2\sqrt{4 - k^2} = 2\sqrt{3} \Rightarrow k^2 = 1$.

Case $k = 1$: $3h - 4 - 5 = 0 \Rightarrow h = 3$. The distance of the centre from the y-axis is $3 > 2 = r$, so this circle does not meet the y-axis; it is rejected.

Case $k = -1$: $3h + 4 - 5 = 0 \Rightarrow h = \dfrac{1}{3}$. Here $|h| < r$, so the circle meets the y-axis.

$C : \left(x - \frac{1}{3}\right)^2 + (y + 1)^2 = 4$. Its chord on the y-axis has length $2\sqrt{r^2 - h^2} = 2\sqrt{4 - \frac{1}{9}} = 2\sqrt{\frac{35}{9}} = \dfrac{2\sqrt{35}}{3}$.

Correct Option: (A)

Question 03

English

For a real number t, let L_t denote the line $x + ty = t^2$. Let S be the set of all real t for which the points $(1, 1)$ and $(4, 2)$ lie on opposite sides of L_t. Then S is a union of two disjoint open intervals whose total length is

(A) $2\sqrt{5}$
(B) $\sqrt{5}$
(C) $1 + \sqrt{5}$
(D) $\dfrac{\sqrt{5}}{2}$

Only one option is correct.

हिन्दी

एक वास्तविक संख्या t के लिए, माना L_t रेखा $x + ty = t^2$ को दर्शाता है। माना S उन सभी वास्तविक t का समुच्चय है जिनके लिए बिन्दु $(1, 1)$ तथा $(4, 2)$ रेखा L_t के विपरीत ओर स्थित हैं। तब S दो असंयुक्त विवृत अन्तरालों का सम्मिलन है जिनकी कुल लम्बाई है

(A) $2\sqrt{5}$
(B) $\sqrt{5}$
(C) $1 + \sqrt{5}$
(D) $\dfrac{\sqrt{5}}{2}$

केवल एक विकल्प सही है।

Solution

Write $f(x, y) = x + ty - t^2$. Two points lie on opposite sides of $f = 0$ exactly when the values of f at them have opposite signs.

$f(1, 1) = 1 + t - t^2$ and $f(4, 2) = 4 + 2t - t^2$. We need $(1 + t - t^2)(4 + 2t - t^2) < 0$.

Zeros: $t^2 - t - 1 = 0 \Rightarrow t = \dfrac{1 \pm \sqrt{5}}{2} \approx -0.618,\ 1.618$; $t^2 - 2t - 4 = 0 \Rightarrow t = 1 \pm \sqrt{5} \approx -1.236,\ 3.236$.

On the number line: $1 - \sqrt{5} < \dfrac{1 - \sqrt{5}}{2} < \dfrac{1 + \sqrt{5}}{2} < 1 + \sqrt{5}$.

$t < 1 - \sqrt{5}$: both factors negative, product positive.

$1 - \sqrt{5} < t < \dfrac{1 - \sqrt{5}}{2}$: first negative, second positive, product negative.

$\dfrac{1 - \sqrt{5}}{2} < t < \dfrac{1 + \sqrt{5}}{2}$: both positive, product positive.

$\dfrac{1 + \sqrt{5}}{2} < t < 1 + \sqrt{5}$: first negative, second positive, product negative.

$t > 1 + \sqrt{5}$: both negative, product positive.

At the four zeros one of the points lies on L_t, so they are excluded.

$$\text{Total length} = \left(\frac{1 - \sqrt{5}}{2} - (1 - \sqrt{5})\right) + \left(1 + \sqrt{5} - \frac{1 + \sqrt{5}}{2}\right) = \frac{\sqrt{5} - 1}{2} + \frac{\sqrt{5} + 1}{2} = \sqrt{5}.$$

Correct Option: (B)

Question 04

English

A ray of light passing through the point $A(-3,0)$ strikes a plane mirror lying along the line $16x + 15y = 100$ at the point $P\left(4, \frac{12}{5}\right)$ and is reflected. The reflected ray meets the x-axis at Q. Let E be the ellipse with foci A and Q that touches the line $16x + 15y = 100$. Then the eccentricity of E is

(A) $\frac{4}{5}$

(B) $\frac{3}{4}$

(C) $\frac{1}{2}$

(D) $\frac{3}{5}$

Only one option is correct.

हिन्दी

बिन्दु $A(-3,0)$ से होकर जाती एक प्रकाश किरण रेखा $16x+15y = 100$ के अनुदिश रखे समतल दर्पण पर बिन्दु $P\left(4, \frac{12}{5}\right)$ पर टकराकर परावर्तित होती है। परावर्तित किरण x-अक्ष से Q पर मिलती है। माना E वह दीर्घवृत्त है जिसकी नाभियाँ A तथा Q हैं और जो रेखा $16x + 15y = 100$ को स्पर्श करता है। तब E की उत्केन्द्रता है

(A) $\frac{4}{5}$

(B) $\frac{3}{4}$

(C) $\frac{1}{2}$

(D) $\frac{3}{5}$

केवल एक विकल्प सही है।

Solution

By the law of reflection, the reflected ray lies along the image of the incident line AP in the mirror, so it passes through A', the image of A in $16x + 15y - 100 = 0$.

Value at A: $16(-3) + 15 \cdot 0 - 100 = -148$. Image: $A' = \left(-3 - \dfrac{2 \cdot 16(-148)}{481},\ 0 - \dfrac{2 \cdot 15(-148)}{481}\right) = \left(\dfrac{89}{13}, \dfrac{120}{13}\right)$.

Line through $P\left(4, \frac{12}{5}\right)$ and A': slope $= \dfrac{\frac{120}{13} - \frac{12}{5}}{\frac{89}{13} - 4} = \dfrac{444/65}{37/13} = \dfrac{12}{5}$, so $y - \frac{12}{5} = \frac{12}{5}(x - 4)$; at $y = 0$, $x = 3$. So $Q = (3, 0)$.

Reflection property of the ellipse: the normal at any point of an ellipse bisects the angle between the focal radii. For the ellipse with foci A, Q that touches the mirror line, the point of contact is the point where a ray from A reflects through Q, which is P. Hence P lies on E.

So $2a = AP + PQ$. $AP = \sqrt{7^2 + \left(\frac{12}{5}\right)^2} = \sqrt{\frac{1369}{25}} = \dfrac{37}{5}$ and $PQ = \sqrt{1^2 + \left(\frac{12}{5}\right)^2} = \dfrac{13}{5}$, so $2a = 10,\ a = 5$.

$2c = AQ = 6$, so $c = 3$ and $e = \dfrac{c}{a} = \dfrac{3}{5}$.

Check: E is $\dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$; P lies on it ($\frac{16}{25} + \frac{9}{25} = 1$) and its tangent at P is $\frac{4x}{25} + \frac{3y}{20} = 1$, i.e. $16x + 15y = 100$.

Correct Option: (D)

Question 05

English

The coordinate axes are rotated about the origin through an angle θ, $0 < \theta < \dfrac{\pi}{2}$, in the anticlockwise sense. The point whose coordinates are $(2, 3)$ in the original system has coordinates $(3, 2)$ in the new system. Then the equation of the line $y = x$, referred to the new axes X and Y, is

(A) $17Y = 7X$
(B) $7Y = 17X$
(C) $17Y = -7X$
(D) $7Y = -17X$

Only one option is correct.

हिन्दी

निर्देशांक अक्षों को मूल बिन्दु के परितः वामावर्त दिशा में कोण θ, $0 < \theta < \dfrac{\pi}{2}$, से घुमाया जाता है। जिस बिन्दु के मूल निकाय में निर्देशांक $(2, 3)$ हैं, उसके नए निकाय में निर्देशांक $(3, 2)$ हैं। तब नए अक्षों X तथा Y के सापेक्ष रेखा $y = x$ का समीकरण है

(A) $17Y = 7X$
(B) $7Y = 17X$
(C) $17Y = -7X$
(D) $7Y = -17X$

केवल एक विकल्प सही है।

Solution

For an anticlockwise rotation of the axes through θ: $X = x\cos\theta + y\sin\theta$, $Y = -x\sin\theta + y\cos\theta$, and conversely $x = X\cos\theta - Y\sin\theta$, $y = X\sin\theta + Y\cos\theta$.

Putting $(x, y) = (2, 3)$ and $(X, Y) = (3, 2)$: $2\cos\theta + 3\sin\theta = 3$ and $-2\sin\theta + 3\cos\theta = 2$.

Multiply the first by 2 and the second by 3 and add: $13\cos\theta = 12$, so $\cos\theta = \dfrac{12}{13}$; then $3\sin\theta = 3 - \dfrac{24}{13} = \dfrac{15}{13}$, so $\sin\theta = \dfrac{5}{13}$. Both are positive, consistent with $0 < \theta < \frac{\pi}{2}$.

Hence $x = \dfrac{12X - 5Y}{13}$ and $y = \dfrac{5X + 12Y}{13}$.

Substitute in $y = x$: $5X + 12Y = 12X - 5Y$, i.e. $17Y = 7X$.

Check: the line $y = x$ makes 45° with the old x-axis, so in the new system its slope is $\tan(45^\circ - \theta) = \dfrac{1 - \frac{5}{12}}{1 + \frac{5}{12}} = \dfrac{7}{17}$.

Correct Option: (A)

Question 06

English

Let $L_1 : 4x - 3y + 2 = 0$ and $L_2 : 12x + 5y - 22 = 0$. A circle C of radius 7 touches both L_1 and L_2, its centre lies on the bisector of the acute angle between L_1 and L_2, and C and the origin lie on opposite sides of L_2. Then the centre of C is

(A) $\left(-\frac{3}{4}, -12\right)$

(B) $\left(\frac{11}{4}, 16\right)$

(C) $(9, 1)$

(D) $(-7, 3)$

Only one option is correct.

हिन्दी

माना $L_1 : 4x - 3y + 2 = 0$ तथा $L_2 : 12x + 5y - 22 = 0$ है। त्रिज्या 7 का एक वृत्त C दोनों रेखाओं L_1 तथा L_2 को स्पर्श करता है, इसका केन्द्र L_1 तथा L_2 के बीच के न्यून कोण के समद्विभाजक पर स्थित है, तथा C और मूल बिन्दु L_2 के विपरीत ओर स्थित हैं। तब C का केन्द्र है

(A) $\left(-\frac{3}{4}, -12\right)$

(B) $\left(\frac{11}{4}, 16\right)$

(C) $(9, 1)$

(D) $(-7, 3)$

केवल एक विकल्प सही है।

Solution

Write both lines with positive constant terms: $L_1 : 4x - 3y + 2 = 0$ and $L_2 : -12x - 5y + 22 = 0$.

Here $a_1a_2 + b_1b_2 = 4(-12) + (-3)(-5) = -33 < 0$. So the origin lies in the acute angle between the lines, and the bisector of the angle containing the origin, $\dfrac{4x - 3y + 2}{5} = +\dfrac{-12x - 5y + 22}{13}$, is the acute-angle bisector.

Simplifying: $52x - 39y + 26 = -60x - 25y + 110 \Rightarrow 112x - 14y - 84 = 0 \Rightarrow 8x - y - 6 = 0$. (The other bisector, $x + 8y - 17 = 0$, bisects the obtuse angle.)

The centre lies on $8x - y - 6 = 0$, say $(h, 8h - 6)$, and is at distance 7 from L_1 (it is then automatically at distance 7 from L_2):

$\dfrac{|4h - 3(8h - 6) + 2|}{5} = 7 \Rightarrow |20 - 20h| = 35 \Rightarrow h = 1 \mp \dfrac{7}{4}$, i.e. $h = -\dfrac{3}{4}$ or $h = \dfrac{11}{4}$.

The two candidates are $\left(-\frac{3}{4}, -12\right)$ and $\left(\frac{11}{4}, 16\right)$, one in each of the two vertically opposite acute angles.

A circle touching L_2 lies on the same side of L_2 as its centre. $12x + 5y - 22$ is -22 at O, 91 at $\left(\frac{11}{4}, 16\right)$ and -91 at $\left(-\frac{3}{4}, -12\right)$.

So the circle on the opposite side of L_2 from O has centre $\left(\dfrac{11}{4}, 16\right)$. (The points $(9, 1)$ and $(-7, 3)$ are the centres on the obtuse-angle bisector.)

Correct Option: (B)

Question 07

English

Let α be a real number such that the equation $2x^2+5xy+2y^2+5x+\alpha y+2=0$ represents a pair of straight lines that intersect at a point of the y-axis. Then the area of the triangle formed by these two lines and the x-axis is

(A) $\frac{3}{2}$
(B) 1
(C) $\frac{9}{8}$
(D) $\frac{3}{4}$

Only one option is correct.

हिन्दी

माना α एक वास्तविक संख्या है जिसके लिए समीकरण $2x^2+5xy+2y^2+5x+\alpha y+2=0$ सरल रेखाओं के एक युग्म को निरूपित करता है जो y-अक्ष के एक बिन्दु पर प्रतिच्छेद करती हैं। तब इन दोनों रेखाओं तथा x-अक्ष से बने त्रिभुज का क्षेत्रफल है

(A) $\frac{3}{2}$
(B) 1
(C) $\frac{9}{8}$
(D) $\frac{3}{4}$

केवल एक विकल्प सही है।

Solution

The second-degree terms factor as $2x^2+5xy+2y^2=(2x+y)(x+2y)$, so the two lines have slopes -2 and $-\frac{1}{2}$ and can be written as $(2x+y+p)(x+2y+q)=0$.

Expanding: $2x^2+5xy+2y^2+(2q+p)x+(q+2p)y+pq=0$. Comparing with the given equation: $2q+p=5$, $pq=2$, $q+2p=\alpha$.

From $p=5-2q$: $(5-2q)q=2\Rightarrow 2q^2-5q+2=0\Rightarrow q=2$ or $q=\frac{1}{2}$.

The same values follow from the condition $abc+2fgh-af^2-bg^2-ch^2=0$ with $a=2, h=\frac{5}{2}, b=2, g=\frac{5}{2}, f=\frac{\alpha}{2}, c=2$: it reduces to $2\alpha^2-25\alpha+68=0$, so $\alpha=4$ or $\alpha=\frac{17}{2}$.

Case $q=2, p=1, \alpha=4$: lines $2x+y+1=0$ and $x+2y+2=0$. Their intersection: $y=-1-2x$, $x-2-4x+2=0\Rightarrow x=0$, so $(0,-1)$, on the y-axis.

Case $q=\frac{1}{2}, p=4, \alpha=\frac{17}{2}$: lines $2x+y+4=0$ and $x+2y+\frac{1}{2}=0$ meet at $\left(-\frac{5}{2},1\right)$, not on the y-axis. Rejected.

So $\alpha=4$. The lines cut the x-axis at $\left(-\frac{1}{2},0\right)$ and $(-2,0)$, and meet at $(0,-1)$.

Area $=\frac{1}{2}\times$ base $\times$ height $=\frac{1}{2}\cdot\frac{3}{2}\cdot 1=\frac{3}{4}$.

Correct Option: (D)

Question 08

English

Let $A(-4,3)$ and $B(-10,0)$ be two points. For $\lambda > 0$, $\lambda \neq 1$, let P_λ be the point that divides AB internally in the ratio $\lambda : 1$, and let Q_λ be the harmonic conjugate of P_λ with respect to A and B. For exactly one value of λ, the circle with $P_\lambda Q_\lambda$ as a diameter passes through the origin O. This circle meets the x-axis at O and at the point $(\beta, 0)$. Then the value of β is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A(-4,3)$ तथा $B(-10,0)$ दो बिन्दु हैं। $\lambda > 0$, $\lambda \neq 1$ के लिए, माना P_λ वह बिन्दु है जो AB को $\lambda : 1$ के अनुपात में अन्तः विभाजित करता है, तथा Q_λ, A और B के सापेक्ष P_λ का प्रसंवादी संयुग्मी (harmonic conjugate) है। λ के ठीक एक मान के लिए, $P_\lambda Q_\lambda$ को व्यास मानकर खींचा गया वृत्त मूल बिन्दु O से होकर जाता है। यह वृत्त x-अक्ष से O पर तथा बिन्दु $(\beta, 0)$ पर मिलता है। तब β का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the section formula, $P_\lambda = \dfrac{\lambda B + A}{\lambda + 1}$. The harmonic conjugate of P_λ divides AB externally in the same ratio, so $Q_\lambda = \dfrac{\lambda B - A}{\lambda - 1}$.

A point X lies on the circle with diameter PQ exactly when $\overrightarrow{XP} \cdot \overrightarrow{XQ} = 0$. For $X = O$ this is $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 0$.

$\overrightarrow{OP} \cdot \overrightarrow{OQ} = \dfrac{(\lambda B + A) \cdot (\lambda B - A)}{\lambda^2 - 1} = \dfrac{\lambda^2 |B|^2 - |A|^2}{\lambda^2 - 1}$.

With $|A|^2 = 25$ and $|B|^2 = 100$: $100\lambda^2 = 25 \Rightarrow \lambda = \dfrac{1}{2}$ (the only positive value).

Geometric reading: the circle on $P_\lambda Q_\lambda$ is the locus of points X with $XA : XB = \lambda$, so it passes through O exactly when $\lambda = OA : OB = 5 : 10$.

$P = \dfrac{\frac{1}{2}(-10,0) + (-4,3)}{3/2} = \dfrac{(-9,3)}{3/2} = (-6,2)$ and $Q = \dfrac{\frac{1}{2}(-10,0) - (-4,3)}{-1/2} = \dfrac{(-1,-3)}{-1/2} = (2,6)$.

Diametric form: $(x+6)(x-2) + (y-2)(y-6) = 0$, i.e. $x^2 + y^2 + 4x - 8y - 12 + 12 = 0$, so $x^2 + y^2 + 4x - 8y = 0$ (it passes through O, as required).

On the x-axis: $x^2 + 4x = 0 \Rightarrow x = 0$ or $x = -4$. So $\beta = -4$.

Final Answer: -4

Question 09

English

Let $A(0,0)$ and $B(9,0)$. A point C, not on the x-axis, moves in such a way that $CA = 2\,CB$. The bisectors of the interior and the exterior angles of triangle ABC at C meet the line AB at D and E respectively, and G is the centroid of triangle ABC. Then the maximum value of the area of triangle DEG is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A(0,0)$ तथा $B(9,0)$ है। एक बिन्दु C, जो x-अक्ष पर नहीं है, इस प्रकार गति करता है कि $CA = 2\,CB$ है। त्रिभुज ABC के शीर्ष C पर अन्तः कोण तथा बाह्य कोण के समद्विभाजक रेखा AB से क्रमशः D तथा E पर मिलते हैं, तथा G त्रिभुज ABC का केन्द्रक है। तब त्रिभुज DEG के क्षेत्रफल का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the interior and exterior angle-bisector theorems, D divides AB internally and E divides AB externally in the ratio $CA : CB = 2 : 1$. So D and E are harmonic conjugates with respect to A, B, and they do not depend on the position of C.

Section formula: $D = \dfrac{2(9,0) + 1(0,0)}{3} = (6,0)$ and $E = \dfrac{2(9,0) - 1(0,0)}{2-1} = (18,0)$.

The two bisectors at C are perpendicular, so $\angle DCE = 90^\circ$ and C lies on the circle with diameter DE: centre $(12,0)$, radius 6.

(Directly: $CA^2 = 4CB^2 \Rightarrow x^2 + y^2 = 4\left((x-9)^2 + y^2\right) \Rightarrow x^2 + y^2 - 24x + 108 = 0$, the same circle.)

Write $C = (12 + 6\cos\theta,\, 6\sin\theta)$, $\sin\theta \neq 0$. The centroid is $G = \left(\dfrac{0+9+12+6\cos\theta}{3}, \dfrac{6\sin\theta}{3}\right) = (7 + 2\cos\theta,\, 2\sin\theta)$, so G moves on a circle of radius 2 centred at $(7,0)$.

D and E lie on the x-axis with $DE = 12$, so the area of triangle DEG is $\dfrac{1}{2} \cdot 12 \cdot |2\sin\theta| = 12|\sin\theta|$.

The maximum is 12, attained for $\theta = \pm\dfrac{\pi}{2}$, i.e. $C = (12, \pm 6)$, which is allowed.

Final Answer: 12

Question 10

English

Let $h > 0$ be such that the two lines represented by $3x^2 + 2hxy + 2y^2 = 0$ are inclined to each other at an angle $\frac{\pi}{4}$. These two lines and the line $x + 3y + 2 = 0$ form a triangle whose orthocentre is (α, β). Then the value of β is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $h > 0$ इस प्रकार है कि $3x^2 + 2hxy + 2y^2 = 0$ द्वारा निरूपित दोनों रेखाएँ एक-दूसरे से कोण $\frac{\pi}{4}$ पर झुकी हैं। ये दोनों रेखाएँ तथा रेखा $x + 3y + 2 = 0$ एक त्रिभुज बनाती हैं जिसका लम्बकेन्द्र (α, β) है। तब β का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $ax^2+2hxy+by^2 = 0$ the angle θ between the lines satisfies $\tan\theta = \dfrac{2\sqrt{h^2-ab}}{|a+b|}$. Here $a = 3, b = 2$: $\dfrac{2\sqrt{h^2-6}}{5} = 1 \Rightarrow h^2 - 6 = \dfrac{25}{4} \Rightarrow h^2 = \dfrac{49}{4}$, and $h > 0$ gives $h = \dfrac{7}{2}$.

The pair is $3x^2 + 7xy + 2y^2 = 0$, i.e. $(3x + y)(x + 2y) = 0$: the lines $y = -3x$ and $y = -\dfrac{x}{2}$. (Check: $\tan\theta = \left|\dfrac{-3+\frac{1}{2}}{1+\frac{3}{2}}\right| = 1$.)

Vertices: $O(0,0)$; $y = -3x$ with $x + 3y + 2 = 0$ gives $-8x + 2 = 0$, $P\left(\frac{1}{4}, -\frac{3}{4}\right)$; $y = -\frac{x}{2}$ with $x + 3y + 2 = 0$ gives $-\frac{x}{2} + 2 = 0$, $Q(4, -2)$.

The line PQ is $x + 3y + 2 = 0$ with slope $-\frac{1}{3}$, so the altitude from O is $y = 3x$.

OQ has slope $-\frac{1}{2}$, so the altitude from P has slope 2: $y + \frac{3}{4} = 2\left(x - \frac{1}{4}\right)$, i.e. $y = 2x - \frac{5}{4}$.

Intersection: $3x = 2x - \frac{5}{4} \Rightarrow x = -\frac{5}{4}$, $y = -\frac{15}{4}$.

Check with the third altitude: from Q perpendicular to OP (slope -3) is $y + 2 = \frac{1}{3}(x - 4)$; at $x = -\frac{5}{4}$, $y = -2 - \frac{7}{4} = -\frac{15}{4}$. So the orthocentre is $\left(-\frac{5}{4}, -\frac{15}{4}\right)$ and $\beta = -3.75$.

Final Answer: -3.75

Question 11

English

Let $A(1,2)$, $B(-2,6)$ and $C(7,10)$ be the vertices of triangle ABC, and let S and H be its circumcentre and orthocentre respectively. Let R_1 and R_2 be the circumradii of triangles SBC and HBC respectively. Then the value of $\dfrac{R_2}{R_1}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A(1,2)$, $B(-2,6)$ तथा $C(7,10)$ त्रिभुज ABC के शीर्ष हैं, तथा S और H क्रमशः इसके परिकेन्द्र और लम्बकेन्द्र हैं। माना R_1 तथा R_2 क्रमशः त्रिभुजों SBC तथा HBC की परित्रिज्याएँ हैं। तब $\dfrac{R_2}{R_1}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$AB = |(-3,4)| = 5$, $AC = |(6,8)| = 10$, $BC = |(9,4)| = \sqrt{97}$. $\cos A = \dfrac{\overrightarrow{AB}\cdot\overrightarrow{AC}}{AB\cdot AC} = \dfrac{-18+32}{50} = \dfrac{7}{25}$.

At B: $\overrightarrow{BA}\cdot\overrightarrow{BC} = (3,-4)\cdot(9,4) = 11 > 0$; at C: $\overrightarrow{CA}\cdot\overrightarrow{CB} = (-6,-8)\cdot(-9,-4) = 86 > 0$. So the triangle is acute, S and H lie inside it.

Triangle HBC: in an acute triangle $\angle BHC = 180^\circ - A$. By the sine rule its circumradius is $R_2 = \dfrac{BC}{2\sin(180^\circ - A)} = \dfrac{BC}{2\sin A} = R$, the circumradius of ABC.

Triangle SBC: $SB = SC = R$ and the central angle is $\angle BSC = 2A$. So $R_1 = \dfrac{BC}{2\sin 2A}$. With $BC = 2R\sin A$: $R_1 = \dfrac{2R\sin A}{4\sin A\cos A} = \dfrac{R}{2\cos A}$.

Hence $\dfrac{R_2}{R_1} = 2\cos A = \dfrac{14}{25} = 0.56$.

Numerical check: $S = \left(\frac{37}{12}, \frac{107}{16}\right)$, $H = \left(-\frac{1}{6}, \frac{37}{8}\right)$, $R = R_2 \approx 5.1296$, $R_1 \approx 9.1600$, ratio 0.56.

Final Answer: 0.56

Question 12

English

Let Γ be the circle $x^2 + y^2 = 16900$, and let $P(100, 75)$ and $Q(40, -30)$. Let K_1 be the locus of the midpoints of the chords of Γ that pass through P, and let K_2 be the locus of the feet of the perpendiculars drawn from Q to the lines passing through P. Then the length of the common chord of K_1 and K_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना Γ वृत्त $x^2 + y^2 = 16900$ है, तथा $P(100, 75)$ और $Q(40, -30)$ हैं। माना K_1, Γ की उन जीवाओं के मध्य-बिन्दुओं का बिन्दुपथ है जो P से होकर जाती हैं, तथा K_2, Q से P से होकर जाने वाली रेखाओं पर डाले गए लम्बों के पादों का बिन्दुपथ है। तब K_1 तथा K_2 की उभयनिष्ठ जीवा की लम्बाई _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$100^2 + 75^2 = 15625 < 16900$, so P lies inside Γ and every line through P cuts a chord.

If M is the midpoint of a chord through P, then $OM \perp$ chord, so $\angle OMP = 90^\circ$ and M lies on the circle with diameter OP. Conversely every point of that circle (inside Γ, which holds since the whole circle lies inside Γ) is such a midpoint. $K_1 : x(x-100)+y(y-75) = 0$, i.e. $x^2+y^2-100x-75y = 0$.

If F is the foot of the perpendicular from Q to a line through P, then $\angle QFP = 90^\circ$, so F lies on the circle with diameter PQ. $K_2 : (x-100)(x-40) + (y-75)(y+30) = 0$, i.e. $x^2 + y^2 - 140x - 45y + 4000 - 2250 = 0$, so $x^2 + y^2 - 140x - 45y + 1750 = 0$.

Common chord: subtract, $40x - 30y - 1750 = 0$, i.e. $4x - 3y - 175 = 0$.

K_1 has centre $(50, 37.5)$ and radius $\frac{1}{2}OP = 62.5$. Its distance from the chord is $\dfrac{|4(50) - 3(37.5) - 175|}{5} = \dfrac{87.5}{5} = 17.5$.

Half chord $= \sqrt{62.5^2 - 17.5^2} = \sqrt{3906.25 - 306.25} = \sqrt{3600} = 60$; the length is 120.

Check: the two circles meet at $P(100, 75)$ and at $(28, -21)$, the foot of the perpendicular from P on OQ (both right angles at that point make O, it and Q collinear); the distance between them is $\sqrt{72^2 + 96^2} = 120$.

Final Answer: 120

Question 13

English

A circle touches the line $4x - 3y = 0$ and cuts off a chord of length 3 on the x-axis. The locus of the centre of such a circle is a conic $\mathcal{C}$. Then the square of the distance between the foci of $\mathcal{C}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक वृत्त रेखा $4x - 3y = 0$ को स्पर्श करता है तथा x-अक्ष पर 3 लम्बाई की जीवा काटता है। ऐसे वृत्त के केन्द्र का बिन्दुपथ एक शांकव $\mathcal{C}$ है। तब $\mathcal{C}$ की नाभियों के बीच की दूरी का वर्ग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let the centre be (h, k) and the radius r. Touching $4x - 3y = 0$: $r = \dfrac{|4h - 3k|}{5}$. A chord of length 3 on the x-axis: $r^2 - k^2 = \left(\dfrac{3}{2}\right)^2 = \dfrac{9}{4}$.

Eliminating r: $\dfrac{(4h - 3k)^2}{25} - k^2 = \dfrac{9}{4}$. Multiply by 25: $16h^2 - 24hk + 9k^2 - 25k^2 = \dfrac{225}{4}$, so the locus is $16x^2 - 24xy - 16y^2 - \dfrac{225}{4} = 0$.

Identify the conic: $a = 16,\ h = -12,\ b = -16, c = -\dfrac{225}{4}$, $g = f = 0$. $\Delta = abc - ch^2 = c(ab - h^2) = -\dfrac{225}{4}(-256 - 144) \neq 0$, so it is not a pair of lines; $h^2 - ab = 400 > 0$, so it is a hyperbola; and $a + b = 0$, so it is a rectangular hyperbola with $e = \sqrt{2}$.

Its centre is the origin (no first-degree terms). Rotating the axes to remove the xy term, the new coefficients are the eigenvalues of $\begin{pmatrix} 16 & -12 \\ -12 & -16 \end{pmatrix}$: $\lambda^2 - 400 = 0$, $\lambda = \pm 20$.

So $20X^2 - 20Y^2 = \dfrac{225}{4}$, i.e. $\dfrac{X^2}{225/80} - \dfrac{Y^2}{225/80} = 1$, with $a^2 = b^2 = \dfrac{225}{80} = \dfrac{45}{16}$.

Distance between the foci $= 2ae$, so its square is $4a^2e^2 = 4 \cdot \dfrac{45}{16} \cdot 2 = \dfrac{45}{2} = 22.5$.

Remark: $d_1^2 - d_2^2$ with d_1, d_2 the distances from two lines always has $a + b = 0$, which is why the locus is rectangular.

Final Answer: 22.5

Question 14

English

Let L_1 and L_2 be the lines represented by $6x^2 - 5xy + y^2 = 0$, where L_1 has the greater slope. A line L through the origin, different from L_2, is inclined to L_1 at the same acute angle at which L_2 is inclined to L_1. Then the slope of L is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना L_1 तथा L_2, $6x^2 - 5xy + y^2 = 0$ द्वारा निरूपित रेखाएँ हैं, जहाँ L_1 की प्रवणता अधिक है। मूल बिन्दु से होकर जाने वाली एक रेखा L, जो L_2 से भिन्न है, L_1 से उसी न्यून कोण पर झुकी है जिस कोण पर L_2, L_1 से झुकी है। तब L की प्रवणता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$6x^2 - 5xy + y^2 = 0$ factors as $(3x - y)(2x - y) = 0$, so the lines are $y = 3x$ and $y = 2x$; $L_1 : y = 3x$ (greater slope) and $L_2 : y = 2x$.

The acute angle θ between them: with $a = 6,\ h = -\frac{5}{2},\ b = 1$, $\tan\theta = \dfrac{2\sqrt{h^2 - ab}}{|a + b|} = \dfrac{2\sqrt{\frac{25}{4} - 6}}{7} = \dfrac{1}{7}$ (equivalently $\left|\dfrac{3 - 2}{1 + 6}\right| = \dfrac{1}{7}$).

Let $L : y = mx$. The acute angle between L and L_1 has $\tan = \left|\dfrac{m - 3}{1 + 3m}\right|$, which must equal $\dfrac{1}{7}$.

Case $\dfrac{m - 3}{1 + 3m} = \dfrac{1}{7}$: $7m - 21 = 1 + 3m \Rightarrow m = \dfrac{11}{2}$.

Case $\dfrac{m - 3}{1 + 3m} = -\dfrac{1}{7}$: $7m - 21 = -1 - 3m \Rightarrow m = 2$, which is L_2 itself and is excluded.

A vertical line through O makes with L_1 an angle with $\tan = \dfrac{1}{3} \neq \dfrac{1}{7}$, so it does not qualify. Hence the slope of L is $\dfrac{11}{2} = 5.5$ (L is the reflection of L_2 in L_1).

Final Answer: 5.5

Question 15

English

A line L passes through the point $(1, -7)$ and is inclined at an angle $\frac{\pi}{4}$ to the line $3x - y = 0$. The line L meets the circle $x^2 + y^2 = 20$, and at each point common to L and the circle, the acute angle between L and the tangent to the circle is $\frac{\pi}{3}$. If L meets the x-axis at $(p, 0)$, then the value of p is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक रेखा L बिन्दु $(1, -7)$ से होकर जाती है तथा रेखा $3x - y = 0$ से कोण $\frac{\pi}{4}$ पर झुकी है। रेखा L वृत्त $x^2 + y^2 = 20$ से मिलती है, तथा L और वृत्त के प्रत्येक उभयनिष्ठ बिन्दु पर L तथा वृत्त की स्पर्श रेखा के बीच का न्यून कोण $\frac{\pi}{3}$ है। यदि L, x-अक्ष से $(p, 0)$ पर मिलती है, तो p का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let m be the slope of L. The angle with $3x - y = 0$ (slope 3) is $\frac{\pi}{4}$: $\left|\dfrac{m-3}{1+3m}\right| = 1$.

$m - 3 = 1 + 3m \Rightarrow m = -2$; $m - 3 = -(1 + 3m) \Rightarrow m = \dfrac{1}{2}$.

Angle between a line and a circle: at a common point T, the tangent is perpendicular to OT. If the line makes the acute angle ϕ with the tangent, it makes $\frac{\pi}{2} - \phi$ with OT. The foot N of the perpendicular from O to L gives the right triangle ONT with $OT = r$ and $ON = d = r\cos\phi$ (the angle at T between TN and TO is $\frac{\pi}{2} - \phi$).

Here $r = 2\sqrt{5}$ and $\phi = \frac{\pi}{3}$, so $d = \sqrt{5}$. By symmetry the angle is the same at both common points.

Case $m = -2$ through $(1, -7)$: $y + 7 = -2(x - 1)$, i.e. $2x + y + 5 = 0$; $d = \dfrac{5}{\sqrt{5}} = \sqrt{5}$. This works.

Case $m = \frac{1}{2}$: $y + 7 = \frac{1}{2}(x - 1)$, i.e. $x - 2y - 15 = 0$; $d = \dfrac{15}{\sqrt{5}} = 3\sqrt{5} > 2\sqrt{5}$, so this line does not meet the circle. Rejected.

So $L : 2x + y + 5 = 0$, which meets the x-axis at $x = -\dfrac{5}{2}$.

Final Answer: -2.5

Question 16

English

Let M be the line $12x - 5y + 55 = 0$, and let $A = (2, 3)$. A ray starts at A and makes an angle θ with the positive direction of the x-axis, where $\tan\theta = \dfrac{3}{4}$ and $0 < \theta < \dfrac{\pi}{2}$. A point P on this ray, different from A, is at equal distances from A and from the line M. Then the value of AP is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना M रेखा $12x - 5y + 55 = 0$ है, तथा $A = (2, 3)$ है। एक किरण A से प्रारम्भ होकर धनात्मक x-अक्ष की दिशा से कोण θ बनाती है, जहाँ $\tan\theta = \dfrac{3}{4}$ तथा $0 < \theta < \dfrac{\pi}{2}$ है। इस किरण पर एक बिन्दु P, जो A से भिन्न है, A से तथा रेखा M से समान दूरी पर है। तब AP का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $\tan\theta = \frac{3}{4}$ and θ acute, $\cos\theta = \frac{4}{5}$ and $\sin\theta = \frac{3}{5}$. The parametric form of the ray gives every point as $P = (2 + r\cos\theta,\ 3 + r\sin\theta) = \left(2 + \frac{4r}{5},\ 3 + \frac{3r}{5}\right)$ with $r = AP > 0$.

Distance of P from M: $\dfrac{\left|12\left(2 + \frac{4r}{5}\right) - 5\left(3 + \frac{3r}{5}\right) + 55\right|}{\sqrt{144 + 25}} = \dfrac{\left|24 + \frac{48r}{5} - 15 - 3r + 55\right|}{13} = \dfrac{\left|64 + \frac{33r}{5}\right|}{13}$.

For $r > 0$ the expression inside is positive, so the condition is $\dfrac{64 + \frac{33r}{5}}{13} = r$.

$64 + \dfrac{33r}{5} = 13r \Rightarrow 64 = \dfrac{32r}{5} \Rightarrow r = 10.$

Check: $P = (10, 9)$; distance from M is $\dfrac{|120 - 45 + 55|}{13} = \dfrac{130}{13} = 10 = AP$.

Final Answer: 10

Question 17

English

Let S be the set of all points (x, y), where x and y are positive integers, that lie on the side of the line $3x + 4y = 12$ not containing the origin and whose distance from this line is less than 10. Then the number of elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना S उन सभी बिन्दुओं (x, y) का समुच्चय है, जहाँ x तथा y धनात्मक पूर्णांक हैं, जो रेखा $3x + 4y = 12$ के उस ओर स्थित हैं जिस ओर मूल बिन्दु नहीं है, तथा जिनकी इस रेखा से दूरी 10 से कम है। तब S के अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $f(x, y) = 3x + 4y - 12$. At the origin $f = -12 < 0$, so the side not containing the origin is $f > 0$, i.e. $3x + 4y > 12$.

For such a point the distance from the line is $\dfrac{3x + 4y - 12}{5}$, and it is less than 10 when $3x + 4y < 62$.

So we count positive integers x, y with $13 \le 3x + 4y \le 61$.

Upper bound for y: $y \le \left\lfloor \dfrac{61 - 3x}{4} \right\rfloor$, and $x \le 19$ (since $y \ge 1$ needs $3x \le 57$).

Lower bound: $y \ge 1$, and also $4y \ge 13 - 3x$, which matters only for $x = 1$ ($y \ge 3$) and $x = 2$ ($y \ge 2$).

$x = 1$: $3 \le y \le 14$, 12 values. $x = 2$: $2 \le y \le 13$, 12. $x = 3$: $1 \le y \le 13$, 13. $x = 4$: $y \le 12$, 12. $x = 5$: 11. $x = 6$: 10. $x = 7$: 10. $x = 8$: 9. $x = 9$: 8. $x = 10$: 7. $x = 11$: 7. $x = 12$: 6. $x = 13$: 5. $x = 14$: 4. $x = 15$: 4. $x = 16$: 3. $x = 17$: 2. $x = 18$: 1. $x = 19$: 1.

Sum: $12 + 12 + 13 + 12 + 11 + 10 + 10 + 9 + 8 + 7 + 7 + 6 + 5 + 4 + 4 + 3 + 2 + 1 + 1 = 137$.

Final Answer: 137

Question 18

English

Let S be the circle $x^2 + y^2 - 4x - 8y + 12 = 0$. For a real number m, let S_m be the image of S in the line $y = mx$. Then the sum of all real values of m for which S_m touches the x-axis is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना S वृत्त $x^2 + y^2 - 4x - 8y + 12 = 0$ है। एक वास्तविक संख्या m के लिए, माना S_m रेखा $y = mx$ में S का प्रतिबिम्ब है। तब m के उन सभी वास्तविक मानों का योग, जिनके लिए S_m, x-अक्ष को स्पर्श करता है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

S has centre $C(2, 4)$ and radius $r = \sqrt{2^2 + 4^2 - 12} = 2\sqrt{2}$. Reflection preserves radii, so S_m is the circle of radius $2\sqrt{2}$ whose centre C' is the image of C in the line $mx - y = 0$.

Image formula: $\dfrac{x' - 2}{m} = \dfrac{y' - 4}{-1} = -\dfrac{2(2m - 4)}{m^2 + 1}$. So $y' = 4 + \dfrac{2(2m - 4)}{m^2 + 1} = \dfrac{4m^2 + 4 + 4m - 8}{m^2 + 1} = \dfrac{4m^2 + 4m - 4}{m^2 + 1}$.

S_m touches the x-axis exactly when the distance of C' from the x-axis equals the radius: $\left|\dfrac{4m^2 + 4m - 4}{m^2 + 1}\right| = 2\sqrt{2}$.

Case $+$: $4m^2 + 4m - 4 = 2\sqrt{2}(m^2 + 1) \Rightarrow (4 - 2\sqrt{2})m^2 + 4m - (4 + 2\sqrt{2}) = 0$. The product of the roots is $-\dfrac{4 + 2\sqrt{2}}{4 - 2\sqrt{2}} < 0$, so there are two distinct real roots; their sum is $-\dfrac{4}{4 - 2\sqrt{2}}$.

Case $-$: $4m^2 + 4m - 4 = -2\sqrt{2}(m^2 + 1) \Rightarrow (4 + 2\sqrt{2})m^2 + 4m - (4 - 2\sqrt{2}) = 0$. The product of the roots is $-\dfrac{4 - 2\sqrt{2}}{4 + 2\sqrt{2}} < 0$, so again two real roots; their sum is $-\dfrac{4}{4 + 2\sqrt{2}}$.

The two quadratics have no common root (a common root would satisfy $4m^2 + 4m - 4 = 0$ and $m^2 + 1 = 0$ together). So there are four values of m.

Sum $= -4\left(\dfrac{1}{4 - 2\sqrt{2}} + \dfrac{1}{4 + 2\sqrt{2}}\right) = -4 \cdot \dfrac{8}{16 - 8} = -4$.

Final Answer: -4

Question 19

English

Let $A(1,2)$ and $B(2,1)$. Let P be a point on the x-axis and Q a point on the line $x+y=8$ such that $AP+PQ+QB$ is minimum. Then the slope of the line QB is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A(1,2)$ तथा $B(2,1)$ है। माना P, x-अक्ष पर एक बिन्दु तथा Q, रेखा $x+y=8$ पर एक बिन्दु इस प्रकार हैं कि $AP+PQ+QB$ न्यूनतम है। तब रेखा QB की प्रवणता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A and B lie above the x-axis and on the origin side of $x+y=8$ ($1+2<8$, $2+1<8$).

Let A_1 be the image of A in the x-axis and B_2 the image of B in $x+y=8$: $A_1=(1,-2)$ and, since the image of (a,b) in $x+y=8$ is $(8-b,8-a)$, $B_2=(7,6)$.

For every P on the x-axis $AP=A_1P$, and for every Q on $x+y=8$ $QB=QB_2$. So $AP+PQ+QB=A_1P+PQ+QB_2\geq A_1B_2=\sqrt{36+64}=10$, with equality exactly when P and Q lie on the segment A_1B_2 in the order A_1,P,Q,B_2.

Segment A_1B_2: $(1+6t,\ -2+8t)$, $0\leq t\leq 1$. It meets $y=0$ at $t=\frac{1}{4}$, giving $P\left(\frac{5}{2},0\right)$, and $x+y=8$ at $-1+14t=8$, $t=\frac{9}{14}$, giving $Q\left(\frac{34}{7},\frac{22}{7}\right)$. As $0<\frac{1}{4}<\frac{9}{14}<1$, the minimum is attained there.

This is also the path of a light ray from A reflected first by a mirror along the x-axis and then by a mirror along $x+y=8$: line PQ is the image of line AP in the x-axis, and line QB is the image of line PQ in $x+y=8$.

Line PQ has slope $\frac{8}{6}=\frac{4}{3}$. Reflection in a line of slope -1 sends a direction of slope s to slope $\frac{1}{s}$, so QB has slope $\frac{3}{4}$.

Direct check: $\dfrac{1-\frac{22}{7}}{2-\frac{34}{7}}=\dfrac{-\frac{15}{7}}{-\frac{20}{7}}=\dfrac{3}{4}$.

Final Answer: 0.75

Question 20

English

Let Γ be the circle $x^2+y^2=25$. For a real number λ, let L_λ be the line $(2x-y-4)+\lambda(x+y+1)=0$. For every λ for which L_λ does not pass through the origin O, let C_λ be the circle passing through O and through the two points where L_λ meets Γ. All the circles C_λ pass through a fixed point (α,β) different from O. Then the value of $\alpha+\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना Γ वृत्त $x^2+y^2=25$ है। एक वास्तविक संख्या λ के लिए, माना L_λ रेखा $(2x-y-4)+\lambda(x+y+1)=0$ है। λ के प्रत्येक ऐसे मान के लिए जिसके लिए L_λ मूल बिन्दु O से होकर नहीं जाती, माना C_λ वह वृत्त है जो O से तथा उन दो बिन्दुओं से होकर जाता है जहाँ L_λ, Γ से मिलती है। सभी वृत्त C_λ, O से भिन्न एक स्थिर बिन्दु (α,β) से होकर जाते हैं। तब $\alpha+\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every L_λ passes through the intersection of $2x-y-4=0$ and $x+y+1=0$: adding, $3x-3=0$, so the point is $(1,-2)$. It lies inside Γ $(1+4<25)$, so each L_λ meets Γ in two points. L_λ passes through O when $-4+\lambda=0$, i.e. $\lambda=4$, which is excluded.

The circles through the two points common to Γ and L_λ are $x^2+y^2-25+t\,[(2+\lambda)x+(\lambda-1)y+(\lambda-4)]=0$. Passing through O: $-25+t(\lambda-4)=0$, so $t=\dfrac{25}{\lambda-4}$.

Multiplying by $(\lambda-4)$: $(\lambda-4)(x^2+y^2-25)+25\,[(2+\lambda)x+(\lambda-1)y+(\lambda-4)]=0$, which simplifies (the constants cancel) to $\lambda\left(x^2+y^2+25x+25y\right)+\left(-4x^2-4y^2+50x-25y\right)=0$.

So every C_λ belongs to the family of circles through the common points of $x^2+y^2+25x+25y=0$ and $4x^2+4y^2-50x+25y=0$, and a point lies on all C_λ exactly when it lies on both.

Subtracting the second from 4 times the first: $150x+75y=0$, so $y=-2x$. Then $x^2+4x^2+25x-50x=0\Rightarrow 5x(x-5)=0$.

The points are O and $(5,-10)$. Hence $\alpha+\beta=5-10=-5$.

Check by power of a point: $(1,-2)$ has power $1+4-25=-20$ with respect to Γ; on the ray from O through $(1,-2)$ the point R with $PO\cdot PR=20$ is $(1,-2)+4(1,-2)=(5,-10)$.

Final Answer: -5

Question 21

English

The origin is shifted to a point $O'(h, k)$, the axes remaining parallel to the original axes, and the equation $36x^2 + 100y^2 - 72x + 400y + 211 = 0$ of an ellipse is referred to the new axes X and Y. The new equation contains neither a term in X nor a constant term, and the coefficient of Y in it is positive. Let F be the focus of the ellipse whose X-coordinate is positive. Then the slope of the line $O'F$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

मूल बिन्दु को बिन्दु $O'(h, k)$ पर स्थानान्तरित किया जाता है, अक्ष मूल अक्षों के समान्तर रहते हैं, तथा एक दीर्घवृत्त का समीकरण $36x^2 + 100y^2 - 72x + 400y + 211 = 0$ नए अक्षों X तथा Y के सापेक्ष लिखा जाता है। नए समीकरण में न तो X वाला पद है और न ही अचर पद, तथा इसमें Y का गुणांक धनात्मक है। माना F दीर्घवृत्त की वह नाभि है जिसका X-निर्देशांक धनात्मक है। तब रेखा $O'F$ की प्रवणता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $x = X + h$, $y = Y + k$. Then $36(X+h)^2 + 100(Y+k)^2 - 72(X+h) + 400(Y+k) + 211 = 0$ has coefficient of X: $72h - 72$; coefficient of Y: $200k + 400$; constant: $36h^2 + 100k^2 - 72h + 400k + 211$.

No X term: $h = 1$. No constant term: $36 + 100k^2 - 72 + 400k + 211 = 0 \Rightarrow 100k^2 + 400k + 175 = 0 \Rightarrow 4k^2 + 16k + 7 = 0 \Rightarrow k = -\frac{1}{2}$ or $k = -\frac{7}{2}$.

The coefficient of Y is $200k + 400$: $+300$ for $k = -\frac{1}{2}$ and -300 for $k = -\frac{7}{2}$. So $O' = \left(1, -\frac{1}{2}\right)$. (The vanishing constant means O' lies on the ellipse; with no X term it is an end of the vertical axis.)

Completing squares: $36(x-1)^2 + 100(y+2)^2 = 36 + 400 - 211 = 225$, so $\dfrac{(x-1)^2}{25/4} + \dfrac{(y+2)^2}{9/4} = 1$, an ellipse with centre $(1, -2)$, horizontal major axis, $a = \frac{5}{2}$, $b = \frac{3}{2}$, $c = \sqrt{a^2 - b^2} = 2$.

Foci: $(3, -2)$ and $(-1, -2)$. In the new system ($X = x - 1$, $Y = y + \frac{1}{2}$) they are $\left(2, -\frac{3}{2}\right)$ and $\left(-2, -\frac{3}{2}\right)$, so $F = \left(2, -\frac{3}{2}\right)$.

Slope of $O'F = \dfrac{-\frac{3}{2} - 0}{2 - 0} = -\dfrac{3}{4} = -0.75$.

Final Answer: -0.75

Question 22

English

Consider the curve $23x^2 + 72xy + 2y^2 = 1250$. Then the square of the distance between its foci is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्र $23x^2 + 72xy + 2y^2 = 1250$ पर विचार कीजिए। तब इसकी नाभियों के बीच की दूरी का वर्ग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Here $a = 23$, $2h = 72$ so $h = 36$, $b = 2$, $g = f = 0$, $c = -1250$.

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = c(ab - h^2) = -1250(46 - 1296) \neq 0$, so the curve is not a pair of lines; $h^2 - ab = 1296 - 46 = 1250 > 0$, so it is a hyperbola. With no first-degree terms its centre is the origin.

To remove the xy term rotate the axes through θ with $\tan 2\theta = \dfrac{2h}{a-b} = \dfrac{72}{21} = \dfrac{24}{7}$. Taking 2θ acute, $\cos 2\theta = \frac{7}{25}$, so $\cos\theta = \frac{4}{5}$, $\sin\theta = \frac{3}{5}$.

With $x = X\cos\theta - Y\sin\theta$, $y = X\sin\theta + Y\cos\theta$: coefficient of $X^2 = a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta = \frac{23\cdot 16 + 72\cdot 12 + 2\cdot 9}{25} = \frac{1250}{25} = 50$.

Coefficient of $Y^2 = a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta = \frac{207 - 864 + 32}{25} = -25$. (Check: $50 + (-25) = a + b = 25$ and $50(-25) = ab - h^2 = -1250$.)

So $50X^2 - 25Y^2 = 1250$, i.e. $\dfrac{X^2}{25} - \dfrac{Y^2}{50} = 1$, with transverse axis along X, $a^2 = 25$, $b^2 = 50$.

$c^2 = a^2 + b^2 = 75$, the distance between the foci is $2c = 10\sqrt{3}$, and its square is 300.

Final Answer: 300

Question 23

English

Consider all the circles each of which touches the three lines $y = 0$, $4x - 3y = 0$ and $4x + 3y = 24$. Let r_1 and r_2 be the smallest and the largest of the radii of these circles. Then the value of $\frac{r_1}{r_2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

उन सभी वृत्तों पर विचार कीजिए जिनमें से प्रत्येक तीनों रेखाओं $y = 0$, $4x - 3y = 0$ तथा $4x + 3y = 24$ को स्पर्श करता है। माना r_1 तथा r_2 इन वृत्तों की त्रिज्याओं में सबसे छोटी तथा सबसे बड़ी हैं। तब $\frac{r_1}{r_2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The lines meet at $O(0,0)$ ($y = 0$ and $4x - 3y = 0$), $B(6,0)$ ($y = 0$ and $4x + 3y = 24$) and $C(3,4)$. So $OB = 6$, $OC = BC = 5$, area $\Delta = \frac{1}{2} \cdot 6 \cdot 4 = 12$, $s = 8$.

A circle touching all three lines has its centre equidistant from them, so the centre lies on an angle bisector (internal or external) at every vertex. The bisectors meet in exactly four points: the incentre and the three excentres. So there are four circles.

Bisectors at O of $y = 0$ and $4x - 3y = 0$: $y = \pm\dfrac{4x - 3y}{5} \Rightarrow x - 2y = 0$ (internal, the triangle lies in $0 < y < \frac{4x}{3}$) and $2x + y = 0$ (external).

Incentre: on $x - 2y = 0$ and on the axis of symmetry $x = 3$, so $I\left(3, \frac{3}{2}\right)$, $r = \frac{3}{2}$ (also $\frac{\Delta}{s} = \frac{12}{8}$).

Excentre opposite C: on $2x + y = 0$ and $x = 3$: $(3, -6)$, radius $6 = \frac{\Delta}{s-6}$.

Excentre opposite O: on $x - 2y = 0$ at distance from $y = 0$ equal to the distance from $4x + 3y = 24$: $(8, 4)$, radius $4 = \frac{\Delta}{s-5}$; by symmetry the excentre opposite B is $(-2, 4)$, radius 4.

So the radii are $\frac{3}{2}, 4, 4, 6$; $r_1 = \frac{3}{2}$, $r_2 = 6$, and $\dfrac{r_1}{r_2} = \dfrac{1}{4} = 0.25$.

Final Answer: 0.25

Question 24

English

The equation $28x^2+24xy-4y^2-92x-12y+55=0$ represents a pair of straight lines. The bisector of the acute angle between these lines meets the x-axis at $(p,0)$. Then the value of p is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

समीकरण $28x^2+24xy-4y^2-92x-12y+55=0$ सरल रेखाओं के एक युग्म को निरूपित करता है। इन रेखाओं के बीच के न्यून कोण का समद्विभाजक x-अक्ष से $(p,0)$ पर मिलता है। तब p का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The second-degree part factors as $28x^2+24xy-4y^2=4(7x^2+6xy-y^2)=4(7x-y)(x+y)$, so the lines have slopes 7 and -1.

Their point of intersection is where both partial derivatives vanish: $56x+24y-92=0$ and $24x-8y-12=0$. From the second, $y=3x-\frac{3}{2}$; then $56x+72x-36-92=0 \Rightarrow x=1$, $y=\frac{3}{2}$. So $P\left(1,\frac{3}{2}\right)$.

Shifting the origin to P ($x=X+1$, $y=Y+\frac{3}{2}$) removes the first-degree terms and the constant, leaving $28X^2+24XY-4Y^2=0$.

The bisectors of $aX^2+2hXY+bY^2=0$ are $\dfrac{X^2-Y^2}{a-b}=\dfrac{XY}{h}$: $\dfrac{X^2-Y^2}{32}=\dfrac{XY}{12} \Rightarrow 3X^2-8XY-3Y^2=0 \Rightarrow (3X+Y)(X-3Y)=0$.

So the bisectors have slopes -3 and $\frac{1}{3}$. To identify the acute one: the angle between slope 7 and slope -3 has $\tan=\left|\dfrac{7+3}{1-21}\right|=\dfrac{1}{2}$, about 26.6°, so this bisector halves an angle of about 53.1°, the acute angle. The other bisector halves the obtuse angle of about 126.9°.

Cross-check with the separated lines $14x-2y-11=0$ and $2x+2y-5=0$: written with positive constants, $-14x+2y+11=0$ and $-2x-2y+5=0$ give $a_1a_2+b_1b_2=28-4>0$, so the origin lies in the obtuse angle and the acute bisector is $\dfrac{-14x+2y+11}{10\sqrt{2}}=-\dfrac{-2x-2y+5}{2\sqrt{2}}$, i.e. $3x+y-\frac{9}{2}=0$.

The acute bisector is $3(x-1)+\left(y-\frac{3}{2}\right)=0$, i.e. $3x+y=\frac{9}{2}$. At $y=0$: $p=\frac{3}{2}=1.5$.

Final Answer: 1.5

Question 25

English

Let $A(-2,0)$ and $B(2,0)$, and let P_0 be the point of the line $x+y=5$ at which $PA+PB$, for P on this line, is minimum. Let E be the ellipse with foci A and B that passes through P_0. Then the square of the length of the minor axis of E is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A(-2,0)$ तथा $B(2,0)$ है, तथा P_0 रेखा $x+y=5$ का वह बिन्दु है जिस पर, इस रेखा पर स्थित P के लिए, $PA+PB$ न्यूनतम है। माना E वह दीर्घवृत्त है जिसकी नाभियाँ A तथा B हैं और जो P_0 से होकर जाता है। तब E के लघु अक्ष की लम्बाई का वर्ग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$x+y-5$ is -7 at A and -3 at B, so both points are on the same side of the line.

The image of A in $x+y=5$ is $A'=(5-0,\,5-(-2))=(5,7)$. For P on the line $PA=PA'$, so $PA+PB=PA'+PB\geq A'B=\sqrt{3^2+7^2}=\sqrt{58}$, with equality when P lies on segment $A'B$.

Line $A'B$: $(2+3t,\,7t)$; on $x+y=5$: $2+10t=5$, $t=0.3$, so $P_0=(2.9,\,2.1)$ and the minimum is $\sqrt{58}$.

At P_0 the segments P_0A and P_0B make equal angles with the line $x+y=5$. By the reflection property, the ellipse with foci A,B through P_0 has this line as its tangent at P_0 (no point of the line lies inside E, since $PA+PB\geq\sqrt{58}$ on it).

E has $2a=P_0A+P_0B=\sqrt{58}$, so $a^2=\frac{58}{4}=\frac{29}{2}$, and $c=2$. Then $b^2=a^2-c^2=\frac{29}{2}-4=\frac{21}{2}$.

Check with the tangent property $b^2=d_Ad_B$ (product of the perpendiculars from the foci to a tangent): $\frac{7}{\sqrt{2}}\cdot\frac{3}{\sqrt{2}}=\frac{21}{2}$.

The minor axis is $2b$, and $(2b)^2=4b^2=42$.

Final Answer: 42

Question 26

English

Let $A(0,4)$ and $B(5,5)$, and let Γ be the graph of $y = |x-2|$, $x \in \mathbb{R}$. The point P of Γ at which $PA + PB$ is minimum has abscissa γ. Then the value of γ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A(0,4)$ तथा $B(5,5)$ है, तथा Γ, $y = |x-2|$, $x \in \mathbb{R}$ का आलेख है। Γ का वह बिन्दु P जिस पर $PA + PB$ न्यूनतम है, उसका भुज γ है। तब γ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Γ consists of the ray $y = x - 2$, $x \geq 2$, and the ray $y = 2 - x$, $x \leq 2$, meeting at $V(2,0)$. A and B satisfy $y > |x-2|$, so they lie above Γ and on the same side of each of the two full lines.

On each ray minimise separately by reflection, then compare.

Right ray, line $x - y - 2 = 0$: the image of (a,b) is $(b+2, a-2)$, so $A' = (6,-2)$. $PA + PB = PA' + PB \geq A'B = \sqrt{1^2+7^2} = \sqrt{50}$. Segment $A'B$: $(6-t, -2+7t)$ meets $x - y = 2$ when $8 - 8t = 2$, $t = \frac{3}{4}$, at $\left(\frac{21}{4}, \frac{13}{4}\right)$, which has $x \geq 2$, so the bound is attained on the ray.

Left ray, line $x + y - 2 = 0$: the image of (a,b) is $(2-b, 2-a)$, so $B' = (-3,-3)$ and the bound is $AB' = \sqrt{9+49} = \sqrt{58}$, attained at $(-0.6, 2.6)$ on the ray.

At the corner V: $VA + VB = \sqrt{20} + \sqrt{34} \approx 10.3$.

The least value is $\sqrt{50}$, attained only at $P = \left(\frac{21}{4}, \frac{13}{4}\right)$. Hence $\gamma = \frac{21}{4} = 5.25$.

Final Answer: 5.25

Question 27

English

Let λ and μ be real numbers such that the equation $3x^2 + 5xy - 2y^2 + x + \lambda y + \mu = 0$ represents a pair of straight lines whose point of intersection lies in the fourth quadrant. The line $x + y = 1$ meets these two lines at the points A and B, and $\angle AOB = \dfrac{\pi}{2}$, where O is the origin. Then the value of $\lambda + \mu$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना λ तथा μ वास्तविक संख्याएँ इस प्रकार हैं कि समीकरण $3x^2 + 5xy - 2y^2 + x + \lambda y + \mu = 0$ सरल रेखाओं के एक युग्म को निरूपित करता है जिनका प्रतिच्छेदन बिन्दु चतुर्थ चतुर्थांश में स्थित है। रेखा $x + y = 1$ इन दोनों रेखाओं से बिन्दुओं A तथा B पर मिलती है, तथा $\angle AOB = \dfrac{\pi}{2}$ है, जहाँ O मूल बिन्दु है। तब $\lambda + \mu$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Joint equation of OA and OB: make the given equation homogeneous using $x + y = 1$:

$3x^2 + 5xy - 2y^2 + (x + \lambda y)(x + y) + \mu(x + y)^2 = 0.$

Coefficient of x^2: $3 + 1 + \mu$; coefficient of y^2: $-2 + \lambda + \mu$. Lines through O are perpendicular exactly when these add to zero: $\lambda + 2\mu + 2 = 0$.

The second-degree part factors as $(3x - y)(x + 2y)$, so the component slopes are 3 and $-\frac{1}{2}$ and the lines are $(3x - y + p)(x + 2y + q) = 0$. Expanding: x-coefficient $3q + p = 1$, y-coefficient $2p - q = \lambda$, constant $pq = \mu$.

(Equivalently $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ with $a = 3, h = \frac{5}{2}, b = -2, g = \frac{1}{2}, f = \frac{\lambda}{2}, c = \mu$.)

Put $p = 1 - 3q$: $\lambda = 2 - 7q$ and $\mu = q - 3q^2$. Then $\lambda + 2\mu + 2 = 0$ gives $2 - 7q + 2q - 6q^2 + 2 = 0$, i.e. $6q^2 + 5q - 4 = 0$, so $q = \frac{1}{2}$ or $q = -\frac{4}{3}$.

$q = \frac{1}{2}$: $p = -\frac{1}{2}$, $\lambda = -\frac{3}{2}$, $\mu = -\frac{1}{4}$; lines $3x - y - \frac{1}{2} = 0$ and $x + 2y + \frac{1}{2} = 0$ meet at $\left(\frac{1}{14}, -\frac{2}{7}\right)$, in the fourth quadrant.

$q = -\frac{4}{3}$: $p = 5$, $\lambda = \frac{34}{3}$, $\mu = -\frac{20}{3}$; lines $3x - y + 5 = 0$ and $x + 2y - \frac{4}{3} = 0$ meet at $\left(-\frac{26}{21}, \frac{9}{7}\right)$, in the second quadrant. Rejected.

Check for the chosen pair: $A = \left(\frac{3}{8}, \frac{5}{8}\right)$ and $B = \left(\frac{5}{2}, -\frac{3}{2}\right)$ on $x + y = 1$; $\overrightarrow{OA} \cdot \overrightarrow{OB} = \frac{15}{16} - \frac{15}{16} = 0$. Neither line is parallel to $x + y = 1$.

$\lambda + \mu = -\frac{3}{2} - \frac{1}{4} = -\frac{7}{4} = -1.75$.

Final Answer: -1.75

Question 28

English

Let h and b be real numbers such that the equation $3x^2 + 2hxy + by^2 = 0$ represents a pair of perpendicular straight lines and one of the bisectors of the angles between these lines passes through the point $(1, 2)$. Then the value of $h + b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना h तथा b वास्तविक संख्याएँ इस प्रकार हैं कि समीकरण $3x^2 + 2hxy + by^2 = 0$ परस्पर लम्बवत् सरल रेखाओं के एक युग्म को निरूपित करता है तथा इन रेखाओं के बीच के कोणों का एक समद्विभाजक बिन्दु $(1, 2)$ से होकर जाता है। तब $h + b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$3x^2 + 2hxy + by^2 = 0$ represents perpendicular lines exactly when the sum of the coefficients of x^2 and y^2 is zero: $3 + b = 0$, so $b = -3$. (Then $h^2 - ab = h^2 + 9 > 0$, so the lines are real for every h.)

The bisectors of the angles between the lines $ax^2+2hxy+by^2 = 0$ are $\dfrac{x^2 - y^2}{a - b} = \dfrac{xy}{h}$, i.e. $h(x^2-y^2) = (a - b)xy = 6xy$.

A bisector passes through $(1, 2)$ exactly when $(1, 2)$ satisfies this joint equation: $h(1 - 4) = 6 \cdot 2$, so $-3h = 12$ and $h = -4$. (If $h = 0$ the bisectors are the axes, which do not pass through $(1, 2)$.)

Check: the pair is $3x^2 - 8xy - 3y^2 = (3x + y)(x - 3y) = 0$, slopes -3 and $\frac{1}{3}$, perpendicular; its bisectors are $-4(x^2 - y^2) = 6xy$, i.e. $2x^2 + 3xy - 2y^2 = (2x - y)(x + 2y) = 0$, and $y = 2x$ passes through $(1, 2)$.

$h + b = -4 + (-3) = -7$.

Final Answer: -7

Question 29

English

For an integer k, let $\mathcal{C}_k$ denote the curve $x^2 + kxy + y^2 - 4x - y - 1 = 0$. Here a pair of straight lines is not regarded as a hyperbola. Then the number of integers k with $|k| \leq 10$ for which $\mathcal{C}_k$ is a hyperbola is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक पूर्णांक k के लिए, माना $\mathcal{C}_k$ वक्र $x^2 + kxy + y^2 - 4x - y - 1 = 0$ को दर्शाता है। यहाँ सरल रेखाओं के युग्म को अतिपरवलय नहीं माना गया है। तब उन पूर्णांकों k, $|k| \leq 10$, की संख्या जिनके लिए $\mathcal{C}_k$ एक अतिपरवलय है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Compare with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$: $a = b = 1$, $h = \frac{k}{2}$, $g = -2$, $f = -\frac{1}{2}$, $c = -1$.

Classification of a non-degenerate conic by $h^2 - ab$: ellipse ($e < 1$) if $h^2 < ab$, parabola ($e = 1$) if $h^2 = ab$, hyperbola ($e > 1$) if $h^2 > ab$. Here $h^2 > ab \iff k^2 > 4 \iff |k| \geq 3$ for integers; with $|k| \leq 10$ there are $8 + 8 = 16$ such integers.

The curve is degenerate (a pair of straight lines) when $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$:

$$\Delta = -1 + 2\left(-\tfrac{1}{2}\right)(-2)\tfrac{k}{2} - \tfrac{1}{4} - 4 + \tfrac{k^2}{4} = \frac{k^2 + 4k - 21}{4} = \frac{(k+7)(k-3)}{4}.$$

So $\Delta = 0$ exactly for $k = 3$ and $k = -7$, both with $|k| \geq 3$. For $k = 3$, as a quadratic in x: $x^2 + (3y - 4)x + (y^2 - y - 1) = 0$ has discriminant $5(y - 2)^2$, a perfect square, so the curve is two lines through $(-1, 2)$; for $k = -7$ the discriminant is $5(3y + 2)^2$. These two values are excluded.

For the other 14 integers $\Delta \neq 0$ and $h^2 > ab$, so the curve is a hyperbola (a non-degenerate conic with $h^2 > ab$ always has real points).

Final Answer: 14

Question 30

English

A line L of slope m passes through the point $(0, 2)$ and meets the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ at the points A and B. Let O be the origin. The slopes of the lines OA and OB are equal in magnitude and opposite in sign. Then the value of m is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रवणता m की एक रेखा L बिन्दु $(0, 2)$ से होकर जाती है तथा वृत्त $x^2 + y^2 - 2x - 4y - 4 = 0$ से बिन्दुओं A तथा B पर मिलती है। माना O मूल बिन्दु है। रेखाओं OA तथा OB की प्रवणताएँ परिमाण में समान तथा चिह्न में विपरीत हैं। तब m का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$(0, 2)$ lies inside the circle $(0 + 4 - 0 - 8 - 4 < 0)$, so every line through it meets the circle at two points A, B, neither of which is O (the circle does not pass through O).

$L : y = mx + 2$ can be written $\dfrac{y - mx}{2} = 1$. Making the circle's equation homogeneous with this gives the pair of lines OA, OB:

$$x^2 + y^2 - (2x + 4y) \cdot \frac{y - mx}{2} - 4 \cdot \frac{(y - mx)^2}{4} = 0.$$

Expanding: $x^2 + y^2 - \big(xy - mx^2 + 2y^2 - 2mxy\big) - \big(y^2 - 2mxy + m^2x^2\big) = 0$, i.e. $(1 + m - m^2)x^2 + (4m - 1)xy - 2y^2 = 0$.

Putting $y = kx$: $-2k^2 + (4m - 1)k + (1 + m - m^2) = 0$, whose roots are the slopes k_1, k_2 of OA, OB. (Neither line is vertical, since the coefficient of y^2 is non-zero.)

Equal magnitude and opposite sign means $k_1 + k_2 = 0$ with $k_1 \neq 0$: $\dfrac{4m - 1}{2} = 0$, so $m = \dfrac{1}{4}$.

Check: $k^2 = \dfrac{1 + \frac{1}{4} - \frac{1}{16}}{2} = \dfrac{19}{32}$, so $k = \pm\sqrt{\frac{19}{32}} \approx \pm 0.77$; the points are about $(-1.96, 1.51)$ and $(3.84, 2.96)$.

Final Answer: 0.25

Question 31

English

A line meets the two lines represented by $3x^2 - 7xy + 2y^2 = 0$ at the points A and B, both different from the origin O, such that the centroid of triangle OAB is $(8, 9)$. Then the area of triangle OAB is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक रेखा $3x^2 - 7xy + 2y^2 = 0$ द्वारा निरूपित दोनों रेखाओं से बिन्दुओं A तथा B पर मिलती है, जो दोनों मूल बिन्दु O से भिन्न हैं, इस प्रकार कि त्रिभुज OAB का केन्द्रक $(8, 9)$ है। तब त्रिभुज OAB का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$3x^2 - 7xy + 2y^2 = 0$ factors as $(3x - y)(x - 2y) = 0$, so the lines are $y = 3x$ and $y = \frac{x}{2}$, through O.

Let $A = (s, 3s)$ on $y = 3x$ and $B = (2u, u)$ on $y = \frac{x}{2}$ (the labels can be swapped without changing the triangle).

Centroid of OAB: $\left(\frac{s + 2u}{3}, \frac{3s + u}{3}\right) = (8, 9)$, so $s + 2u = 24$ and $3s + u = 27$.

From the first, $s = 24 - 2u$; then $72 - 6u + u = 27$, so $u = 9$ and $s = 6$. Hence $A = (6, 18)$, $B = (18, 9)$, and the line is $AB : 3x + 4y = 90$.

Area of $OAB = \frac{1}{2}\,|x_A y_B - x_B y_A| = \frac{1}{2}|6 \cdot 9 - 18 \cdot 18| = \frac{1}{2} \cdot 270 = 135$.

Final Answer: 135